

ITU-T G.984.1, Clause 14: Protection on the PON Section

ITU-T G.984.1 第 14 章：PON 段的保护

ITU-T G.984.1 (03/2008) · 原文：itu.int/rec/T-REC-G.984.1

阅读说明：每句英文下方紧跟中文译文（Trancy 式上下排）。本文件仅为学习目的摘录 ITU-T 建议书第 14 章的关键条款，版权归 ITU 所有，完整内容请查阅原文。译文由 AI 生成，准确性以 ITU 原文为准。

14 · 概述 / General

From the viewpoint of administration of the access network, the protection architecture of GPON is considered to enhance the reliability of the access networks.

从接入网管理的角度看，GPON 的保护架构被用来增强接入网的可靠性。

However, protection shall be considered as an optional mechanism because its implementation depends on the realization of economical systems.

然而，保护应被视为一种**可选机制**，因为是否实现取决于系统在经济性上的可行性。

Further information on protection switching can be found in [ITU-T G.808.1].

关于保护倒换的更多信息，可参见 [ITU-T G.808.1]。

This clause presents some possible duplex configurations and related requirements as examples of protected GPON systems.

本章给出若干可能的双份（duplex）配置及相关要求，作为受保护 GPON 系统的示例。

14.1 · 切换类型 / Possible switching types

There are two types of protection switching, both of which are analogous to those of SDH systems: i) automatic switching; and ii) forced switching.

保护倒换有两种类型，都与 SDH 系统类似：i) **自动切换**；ii) **强制切换**。

The first type is triggered by fault detection, such as loss of signal, loss of frame, signal degrade (BER becomes worse than the predetermined threshold), and so on.

第一种由故障检测触发，例如**信号丢失（LOS）、帧丢失（LOF）、信号劣化（BER 超过预定阈值）**等。

The second type is activated by administrative events, such as fibre rerouting, fibre replacement, etc.

第二种由管理性事件激活，例如**光纤改道、光纤更换**等。

Both types should be possible in a GPON system, if required, even though they are optional functions.

如有需要，GPON 系统应能支持这两种切换，尽管它们都是可选功能。

The switching mechanism is generally realized by the OAM function, therefore, the required OAM information field should be reserved in the OAM frame.

切换机制通常由 OAM 功能实现，因此应在 OAM 帧中预留所需的 OAM 信息字段。

14.2.1 • 双份配置示例 / Configuration examples

Type A: A deprecated configuration that duplicated only the optical fibres.

Type A: 一种已废弃的配置，只对光纤做双份。

Type B: The second configuration (Figure 4a) doubles the OLTs and the optical fibres between the OLTs and the optical splitter, and the splitter has two input/output ports on the OLT side.

Type B: 第二种配置（图 4a）对 **OLT** 以及 **OLT 到分光器之间的光纤**做双份，分光器在 OLT 侧有**两个输入/输出端口**。

This configuration reduces the cost of duplexing the ONUs, although only the OLT side can be recovered.

这种配置降低了 ONU 侧双份的成本，但**只能恢复 OLT 侧**。

Type C: The third configuration (Figure 4b) doubles not only the OLT side facilities but also the ONU side.

Type C: 第三种配置（图 4b）不仅对 OLT 侧设施做双份，**也对 ONU 侧做双份**。

In this configuration, recovery from failure at any point is possible by switching to the standby facilities.

在这种配置下，**任意一点**发生故障，都可以通过切换到备用设施来恢复。

Therefore, the full duplex cost enables high reliability.

因此，全双份的代价换来了高可靠性。

Type D: A deprecated configuration that allowed the mixing of duplicated and non-duplicated ONUs, essentially providing a combination of types B and C protection.

Type D: 一种已废弃的配置，允许双份与非双份 ONU 混用，本质上是 Type B 与 Type C 保护的组合。

Note that in both of these types of protection schemes, the PON line terminations do not need to reside in a single OLT equipment.

注意：在这两类保护方案中，PON 线路终端**不必位于同一台 OLT 设备内**。

In fact, they may be located in physically diverse locations (dual parenting).

事实上，它们可以位于物理上分离的位置（即**双归属，dual parenting**）。

14.2.2 • 特性 / Characteristics

Type B: This configuration requires cold standby of the spare circuit in the OLT side.

Type B: 该配置要求 OLT 侧的备用电路处于**冷备**状态。

In this case, signal loss or even frame loss is, in general, inevitable in the switching period.

在这种情况下，切换期间通常不可避免地会发生**信号丢失甚至帧丢失**。

However, all the connections supported between the service node and the terminal equipment should be held after this switching.

不过，切换完成后，业务节点与终端设备之间承载的所有连接都应保持。

Type C: In this case, the hot standby of the spare receiver circuits is possible in both ONU and OLT sides.

Type C：在这种情况下，ONU 侧与 OLT 侧的备用接收电路都可以**热备**。

In addition, hitless switching (without frame loss) is also possible in this configuration.

此外，该配置还可以实现**无损切换（hitless，不丢帧）**。

14.3 • 要求 / Requirements

1) The protection switching function should be optional.

1) 保护倒换功能应为可选。

2) Both automatic protection switching and forced switching are possible in the GPON system, if required, even though they are optional functions.

2) 如有需要，GPON 系统应能同时支持自动保护倒换和强制切换，尽管它们都是可选功能。

保护类型速览 / Protection types at a glance

Type	Duplicated 双份范围	Recoverable 可恢复范围	Switching 切换
A	Fibres only 仅光纤	Deprecated 已废弃	
B	OLT side + fibres to splitter OLT 侧 + 到分光器的光纤	OLT side only 仅 OLT 侧	Cold standby, possible loss 冷备，可能丢信号/丢帧
C	OLT side + ONU side OLT 侧 + ONU 侧	Any point 任意一点	Hot standby, hitless 热备，无损
D	Mix of B and C B 与 C 混合	Deprecated 已废弃	

关键术语 / Key terms: protection switching 保护倒换 • automatic / forced switching 自动/强制切换 • loss of signal (LOS) 信号丢失 • loss of frame (LOF) 帧丢失 • signal degrade 信号劣化 • cold standby 冷备 • hot standby 热备 • hitless switching 无损切换 • duplex 双份 • dual parenting 双归属 • PON LT (line termination) PON 线路终端